1.19 Graph Sketching: Radical Functions

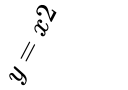
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Transformations of Functions

34

Suppose $f(x) = x^2$

Sketch $y = f(x)$:



y

x

Sketch $y = f(x+2)$



x

What do you notice about the relationship between the graphs of $y = f(x)$ and $y = f(x+2)$?

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Transformations of Functions

35

This is all you need to remember when considering how transforming your function transforms your graph...

	Affects which axis?	What we expect or opposite?
Change **inside** $f()$		
Change **outside** $f()$		

Therefore...

$y = f(x-3)$

$y = f(x) + 4$

$y = f(5x)$

$y = 2f(x)$

1.19 Graph Sketching: Radical Functions

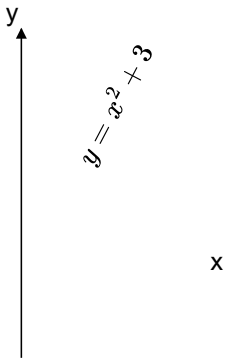
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Sketching transformed graphs

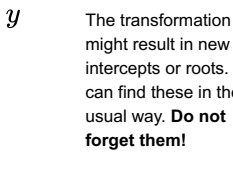
36

Sketch $y = x^2 + 3$

Imagine a sketch of $y = x^2$ and then do the translation, ensuring you adjust any intercepts with the axes.



Sketch $y = \frac{2}{x+1}$



The transformation might result in new intercepts or roots. You can find these in the usual way. **Do not forget them!**

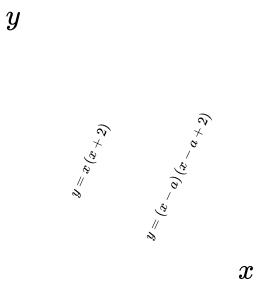
Draw asymptotes using a dotted line and write its equation on it.

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More Examples

37

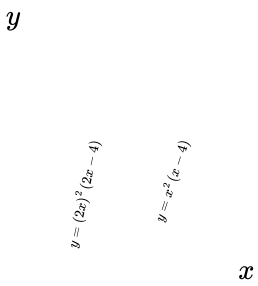
Sketch $y = x(x+2)$. On the same axes, sketch $y = (x-a)(x-a+2)$, where $a > 2$.



Note that our intercepts are in terms of a.

Sketch $y = x^2(x-4)$. On the same axes, sketch the graph with equation

$$y = (2x)^2(2x-4)$$



1.19 Graph Sketching: Radical Functions

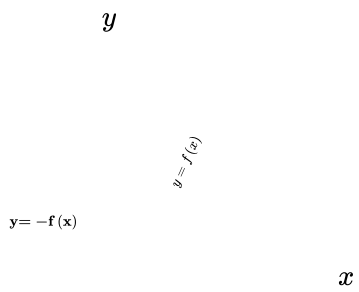
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Reflections of Graphs

38

If $y = x(x+2)$, sketch $y = f(x)$ and $y = -f(x)$ on the same axes.

You did this at GCSE. The minus is outside the function, so affects the output, i.e. the y value. The y values are negated, resulting in a reflection in the x-axis.

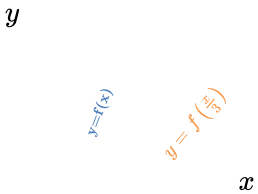


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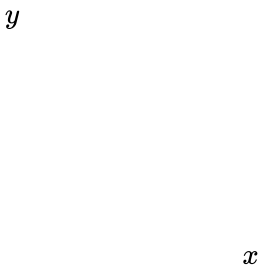
Test Your Understanding

39

If $y = (x+1)(x-2)$, sketch $y = f(x)$ and $y = f(\frac{x}{3})$ on the same axes.



Sketch the graph of $y = \frac{2}{x} + 1$, ensuring you indicate any intercepts with the axes.



1.19 Graph Sketching: Radical Functions

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Exercise 4E – Pure Y1 page 74-75 (translations)

40

7 The point $P(4, -1)$ lies on the curve with equation $y = \mathrm{f}(x)$. **a** State the coordinates that point P is transformed to on the curve with equation $y = \mathrm{f}(x - 2)$. **b** State the coordinates that point P is transformed to on the curve with equation $y = \mathrm{f}(x) + 3$.

8 The graph of $y = \mathrm{f}(x)$ where $\mathrm{f}(x) = \frac{1}{x}$ is translated so that the asymptotes are at $x = 4$ and $y = 0$. Write down the equation for the transformed function in the form $y = \frac{1}{x+a}$.

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Exercise 4E – Pure Y1 page 74-75 (translations)

41

11 a Sketch the graph of $y = x^3 + 4x^2 + 4x$. **b** The poin
coordinates $(-1, 0)$ lies on the curve with equation
$y = (x + a)^3 + 4(x + a)^2 + 4(x + a)$ where a is a constant. Find
the two possible values of a.

1.19 Graph Sketching: Radical Functions

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Exercise 4E – Pure Y1 page 74-75 (translations)

42

12 a Sketch the graph of $y = x(x+1)(x+3)^2$. **b** Find the possible values of b such that the point $(2, 0)$ lies on the curve with equation $y = (x+b)(x+b+1)(x+b+3)^2$.

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Exercise 4F – Pure 1 Page 78 (stretches/reflections)

43

7 The point $P(2, -3)$ lies on the curve with equation $y = f(x)$.

a State the coordinates that point P is transformed to on the curve with equation $y = f(2x)$.

b State the coordinates that point P is transformed to on the curve with equation $y = 4f(x)$.

8 The point $Q(-2, 8)$ lies on the curve with equation $y = f(x)$. State the coordinates that point Q is transformed to on the curve with equation $y = f(\frac{1}{2}x)$.

1.19 Graph Sketching: Radical Functions

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Exercise 4F – Pure 1 Page 78 (stretches/reflections)

44

9 a Sketch the graph of $y = (x-2)(x-3)^2$. **b** The graph of $y = (ax-2)(ax-3)^2$ passes through the point $(1, 0)$. Find two possible values for a.

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Effect of transformation on specific points

45

Sometimes you will not be given the original function, but will be given a sketch with specific points and features you need to transform.
Where would each of these points end up?

$\mathbf{y=f(x)}$	$(\mathbf{4,3})$	$(\mathbf{1,0})$	$(\mathbf{6,-4})$
$y=f(x+1)$			
$y=f(2x)$			
$y=3f(x)$			
$y=f(x)-1$			
$y=f\left(\frac{x}{4}\right)$			
$y=f(-x)$			
$y=-f(x)$			

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Test Your Understanding

46

Edexcel C1 May 2012
Q10

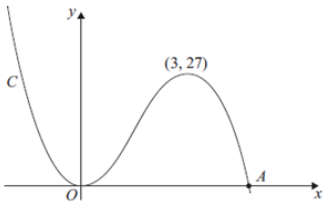


Figure 1

Figure 1 shows a sketch of the curve C with equation $y = f(x)$, where $f(x) = x^2(9 - 2x)$. There is a minimum at the origin, a maximum at the point $(3, 27)$ and C cuts the x-axis at the point A.

(a) Write down the coordinates of the point A. (1)

(b) On separate diagrams sketch the curve with equation
(i) $y = f(x + 3)$,
(ii) $y = 2f(x)$.
On each sketch you should indicate clearly the coordinates of the maximum point and any points where the curves cross or meet the coordinate axes. (5)

The curve with equation $y = f(x) + k$, where k is a constant, has a maximum point at $(3, 10)$.

(c) Write down the value of k. (1)

(a)

(b)(i)

(ii)

(c)

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Exercise 4G Pure Y1 Pages 80

47

5 The point $P(2, 1)$ lies on the graph with equation $y = f(x)$.

a On the graph of $y = f(ax)$, the point P is mapped to the point $Q(4, 1)$. Determine the value of a.

b Write down the coordinates of the point to which P maps under each transformation

i $f(x - 4)$ **ii** $3f(x)$ **iii** $\frac{1}{2}f(x) - 4$

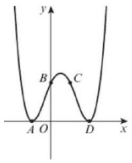
1.19 Graph Sketching: Radical Functions

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Exercise 4G Pure Y1 Pages 80

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6 The diagram shows a sketch of a curve with equation $y = \mathrm{f}(x)$. The points $A(-1, 0), B(0, 2), C(1, 2), D(2, 0)$ lie on the curve. Sketch the following graphs and give the coordinates of the points A, B, C, D after each transformation:



a $y + 2 = \mathrm{f}(x)$ b $\frac{1}{2}y = \mathrm{f}(x)$ c $y - 3 = \mathrm{f}(x)$ d $3y = \mathrm{f}(x)$ e $2y - 1 = \mathrm{f}(x)$